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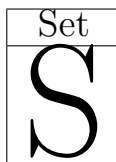
Brac University

Semester: Spring 2026

Course Code: CSE428L

Digital Image Processing Lab

Faculty:



Assessment: *Lab Quiz*

Duration: 40 minutes

Date: April 20, 2026

Full Marks : 25

Instructions:

- All questions are compulsory.
- For MCQs, circle the letter of the best answer.
- Write your answers in the space provided.

■ Question 1 of 4 [10 marks]

Choose the single best answer for each question.

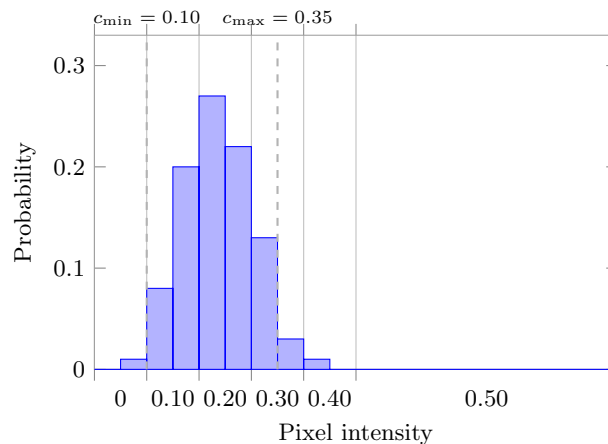
- (a) [1 mark] An image array I has shape (480, 640, 3). What is the shape of $I[:, :2, :]$? A. (480, 320, 3) B. (240, 640, 3) C. (240, 320, 3) D. (120, 160, 3)
- (b) [1 mark] `np.clip(I*2, 0, 1)` is applied to a normalised image where every pixel value lies in $[0, 0.4]$. Which statement is **true**?
- All output values are clamped to 1.
 - The output is identical to $I*2$ — no clipping occurs.
 - All output values are set to 0.
 - The output is identical to I — multiplication is undone.
- (c) [1 mark] A 9×9 kernel is applied to a (100, 100) grayscale image using `mode='valid'`. What is the output shape? A. (100, 100) B. (92, 92) C. (96, 96) D. (108, 108)
- (d) [1 mark] Cross-correlation and convolution of an image with kernel k produce identical results when:
- The image dimensions are square.
 - All values in k are equal.
 - k is symmetric.
 - The `mode` parameter is set to `'same'`.
- (e) [1 mark] A 20 Hz signal is sampled at 30 Hz (below the Nyquist rate). At what frequency does the aliased component appear? A. 10 Hz B. 5 Hz C. 20 Hz D. 50 Hz
- (f) [1 mark] Gamma correction $I_{\text{out}} = I^\gamma$ is applied to a normalised image (pixel values in $[0, 1]$) with $\gamma = 2$. The visual effect is:
- The image is brightened.
 - The image is darkened.
 - Pixel values are inverted around 0.5.
 - Contrast is unchanged; only colour saturation increases.
- (g) [1 mark] A `LinearRegression` model is trained on data where $y \approx 3x^2 + 2$. The model performs very poorly. The most appropriate fix is:
- Engineer a new feature x^2 and retrain the same model.
 - Collect a larger dataset.
 - Replace the model with logistic regression.
 - Decrease the learning rate.
- (h) [1 mark] Three `nn.Linear` layers are stacked with **no activation functions** between them. This model is mathematically equivalent to:
- A deep network capable of non-linear decision boundaries.
 - A single linear transformation.
 - A convolutional neural network.
 - A decision tree.
- (i) [1 mark] In Lab 7's CNN classifier, `nn.Conv2d` layers process the image directly instead of flattening it first. The key advantage is:

- A. Fewer total parameters in all cases.
 - B. Spatial locality is preserved .
 - C. Backpropagation is no longer required.
 - D. Input images are automatically normalised.
- (j) [1 mark] The IoU (Intersection over Union) metric for a **perfect prediction** (every pixel correctly classified) equals: A. 0.0 B. 0.5 C. 1.0 D. Undefined (division by zero)

■ Question 2 of 4 [6 marks]

Answer all questions concisely.

- (a) [2 marks] The histogram below shows the normalised intensity distribution (PDF) of a greyscale satellite image (pixel values in $[0, 1]$).



- (i) What does this histogram reveal about the image's **contrast**? Identify the approximate pixel range where values are concentrated.
 - (ii) Derive the contrast-stretching function that linearly maps $[c_{\min}, c_{\max}] = [0.10, 0.35]$ to $[0, 1]$. Simplify your answer to the form $I_{\text{out}} = m(I_{\text{in}} - b)$.
- (b) [2 marks] In an experiment, each 28×28 MNIST image is flattened to a 784-element vector before being fed to the fully-connected classifier.
- (a) State **one** limitation of this flattening approach.
 - (b) Briefly explain how the convolutional classifier in the same lab overcomes that limitation.
- (c) [2 marks] Write the **Dice Loss** formula (using predicted mask P , ground-truth mask G , and smoothing term ε). In **one sentence**, explain why it is preferred over pixel-wise binary cross-entropy when the lesion region is far smaller than the background.

■ Question 3 of 4 [4 marks]

Read the following code carefully and answer the questions below.

```
1 import numpy as np
2 from skimage import io, color, img_as_float
3 from scipy import signal
4
5 I = io.imread("photo.jpg")
6 I = img_as_float(I)           # Line A
7 I = color.rgb2gray(I)        # Line B
8
9 k = np.ones([9, 9]) / 81.0   # 9x9 box kernel
10
11 I_corr = signal.correlate(I, k, mode='valid') # Line C
12 I_conv = signal.convolve(I, k, mode='valid')  # Line D
13
14 print((I_corr == I_conv).all())              # Line E
```

Assume the image loaded by `io.imread` is a colour JPEG. After **Line B**, `I` has shape (300, 400).

- (a) [1 mark] Before **Line A**, what are the **dtype** and **pixel value range** of `I`?
- (b) [1 mark] After **Line B**, `I` is grayscale. What is its **shape**, and what happened to the channel dimension?
- (c) [1 mark] What is the **shape** of `I_corr` after **Line C**? Show your calculation.
- (d) [1 mark] What does **Line E** print? Briefly justify your answer in terms of the properties of kernel `k`.

■ Question 4 of 4 [5 marks]

Read the following code carefully and answer the questions below.

```
1 import torch
2 import torch.nn as nn
3 import torch.nn.functional as F
4
5 class ImageClassifier(nn.Module):
6     def __init__(self):
7         super().__init__()
8         self.conv1 = nn.Conv2d(in_channels=1, out_channels=16,
9                                 kernel_size=3, padding=1)
10        self.conv2 = nn.Conv2d(in_channels=16, out_channels=32,
11                                kernel_size=3, padding=1)
12        self.pool = nn.MaxPool2d(kernel_size=2, stride=2)
13        self.fc = nn.Linear(32 * 7 * 7, 10)
14
15    def forward(self, x):
16        x = F.relu(self.conv1(x))    # Line A
17        x = self.pool(x)             # Line B
18        x = F.relu(self.conv2(x))    # Line C
19        x = self.pool(x)             # Line D
20        x = x.flatten(1)
21        return self.fc(x)
22
23 x = torch.randn(8, 1, 28, 28)
24 out = ImageClassifier()(x)
```

- (a) [1 mark] What is the shape of **x** after **Line A** and after **Line B**?
- (b) [1 mark] What is the shape of **out**?
- (c) [1 mark] The **fc** layer expects input dimension $32 \times 7 \times 7$, assuming a 28×28 input image. If the input were changed to 56×56 , what should the first argument of **fc** be? Show your reasoning.
- (d) [1 mark] Both **Conv2d** layers use **padding=1** with **kernel_size=3**. What is the spatial effect of this combination on the output feature map?
- (e) [1 mark] What will be the range of values for the **out** variable?